

Conducting a Wi-Fi Site Survey

Wireless and Mobile Device Security, Second Edition - Lab 03

Student:

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Email:



Time on Task:

1 hour, 32 minutes

Progress:

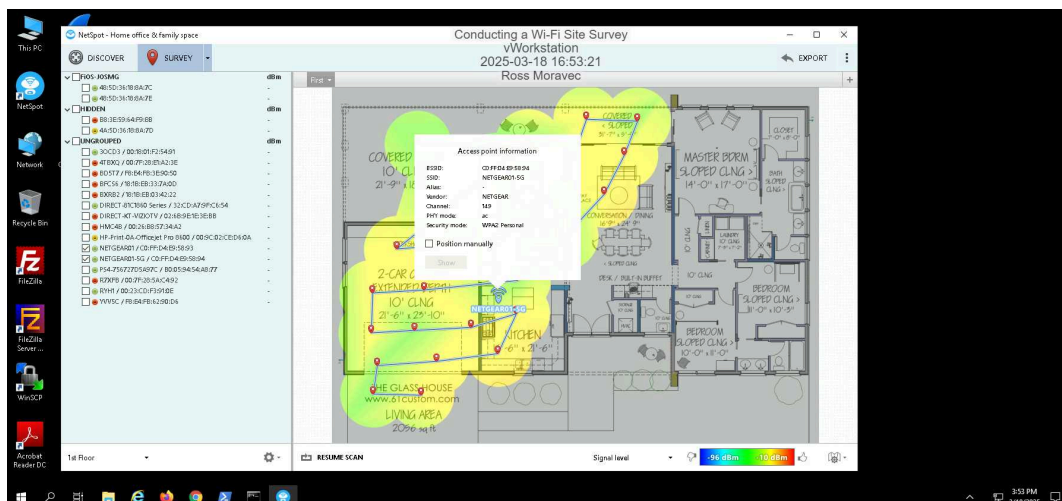
100%

Report Generated: Tuesday, March 18, 2025 at 8:16 PM

Hands-On Demonstration

Part 1: Generate a Heatmap for Signal Level Coverage

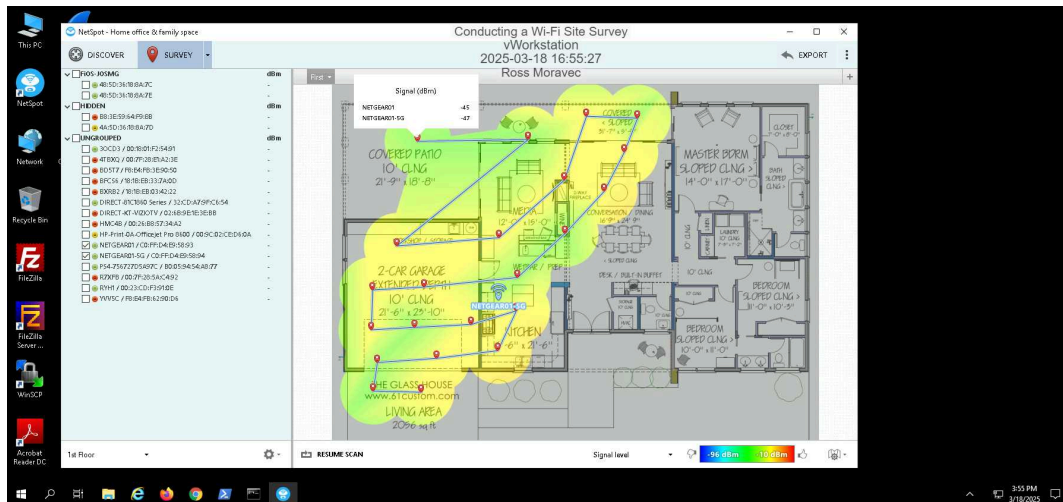
5. Make a screen capture showing the **NETGEAR01-5G** access point details.



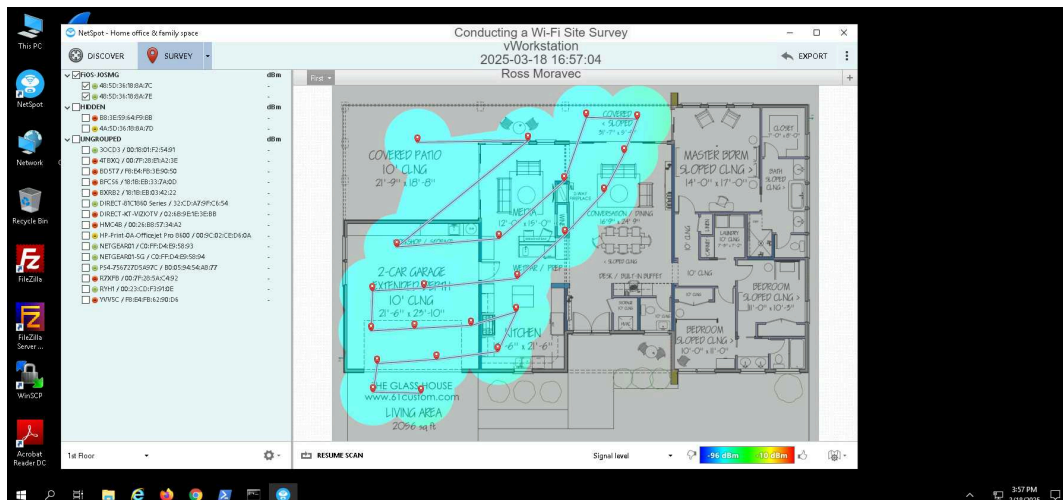
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9. Make a screen capture showing the signal levels recorded at the first sample point.



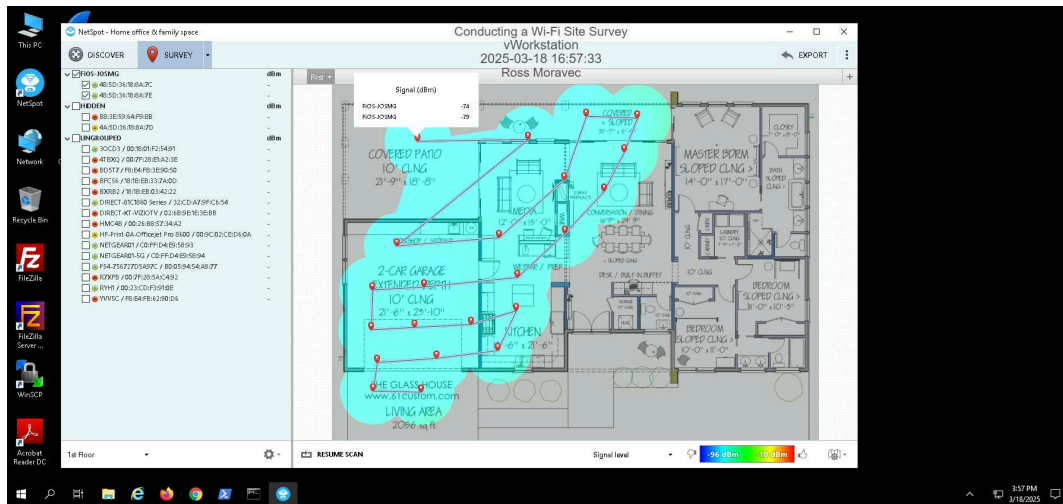
12. Make a screen capture showing the heatmap for the FiOS-JOSMG group only.



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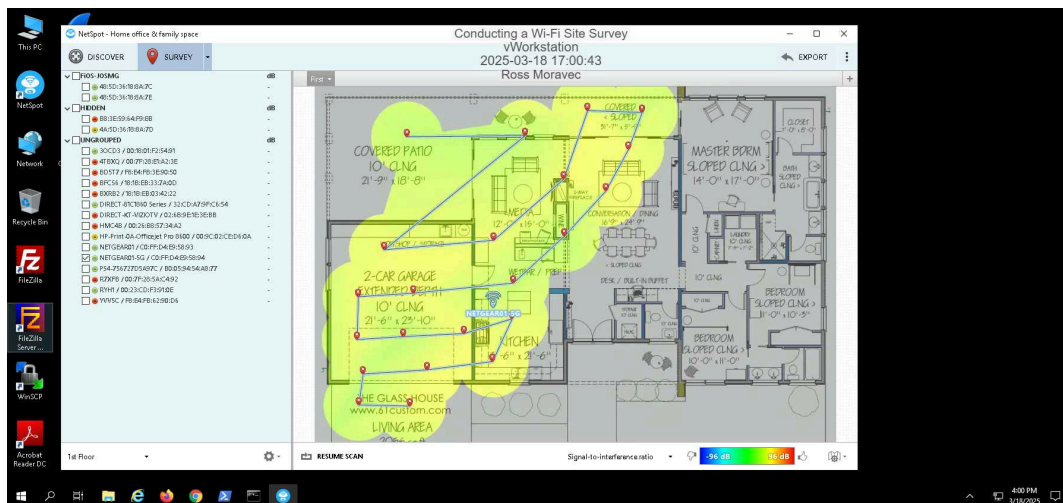
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14. Make a screen capture showing the **FiOS-JOSMG** signal levels detected at the first sample point.



Part 2: Generate Heatmaps for Other Key Attributes

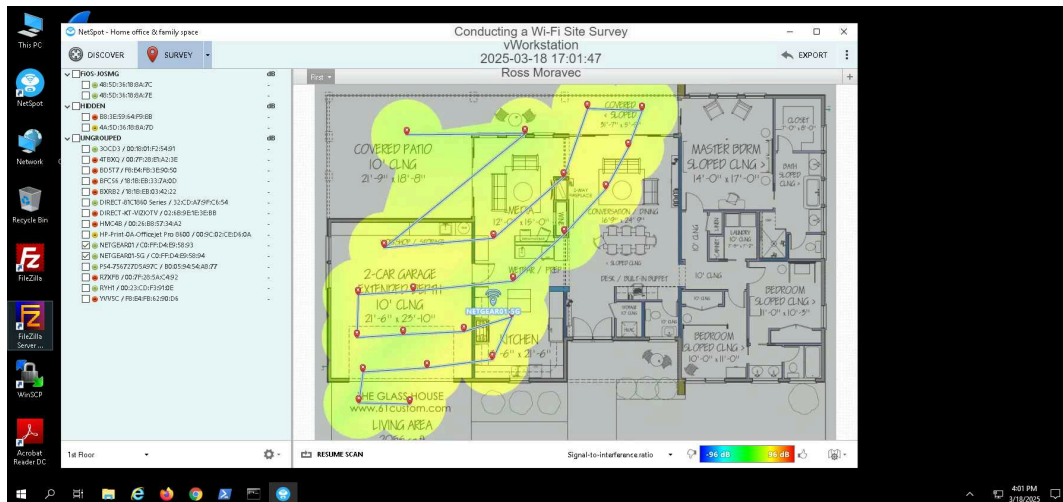
4. Make a screen capture showing the **signal-to-interference ratio** heatmap for the **NETGEAR01-5G** transmitter.



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7. Make a screen capture showing the Netgear SIR levels detected at the first sample point.



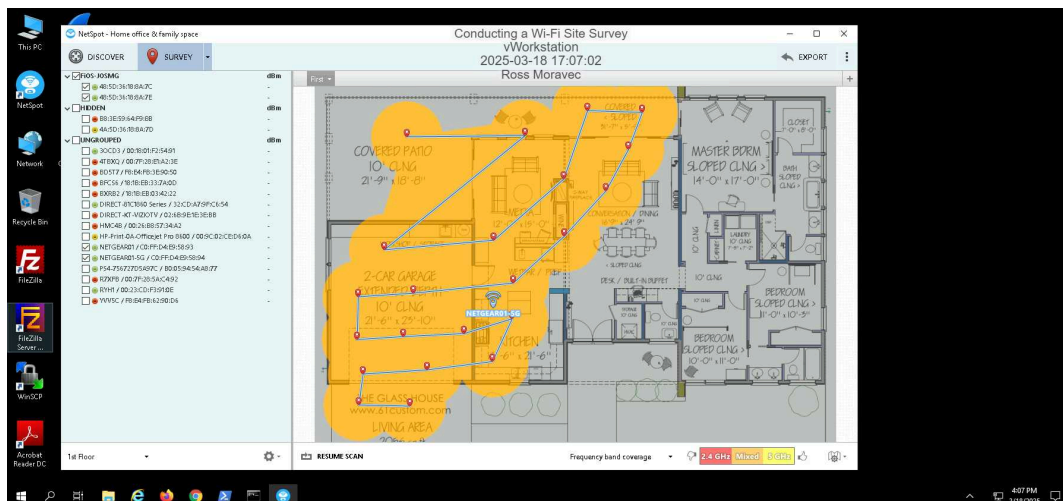
13. Record the PHY mode for the NETGEAR01 transmitter.

n

15. Record the PHY mode for the NETGEAR01-5G transmitter.

ac

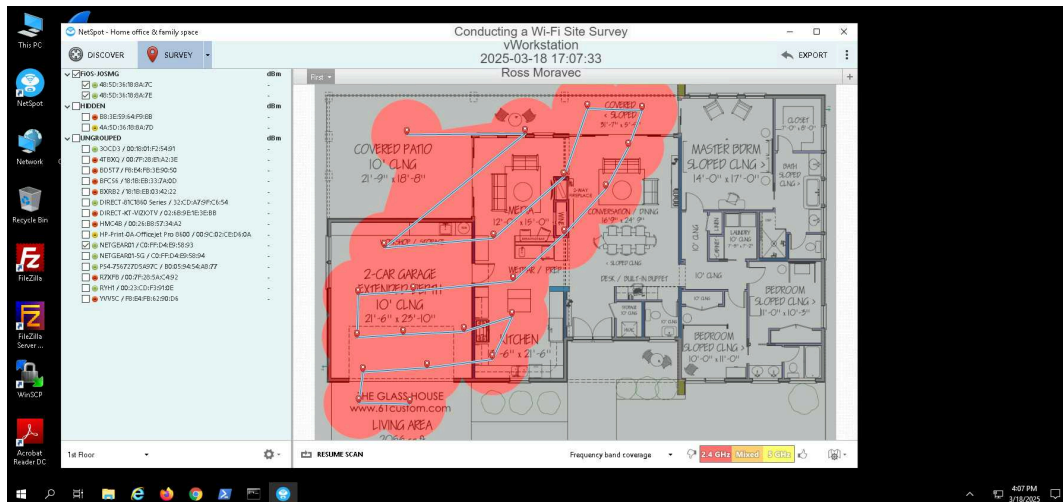
17. Make a screen capture showing the frequency band heatmap.



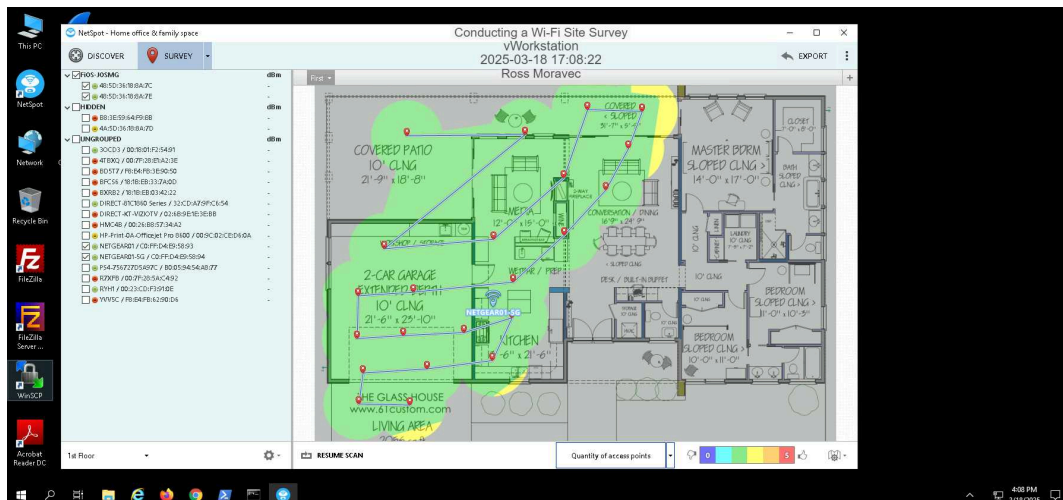
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19. Make a screen capture showing the 2.4 GHz frequency band heatmap.



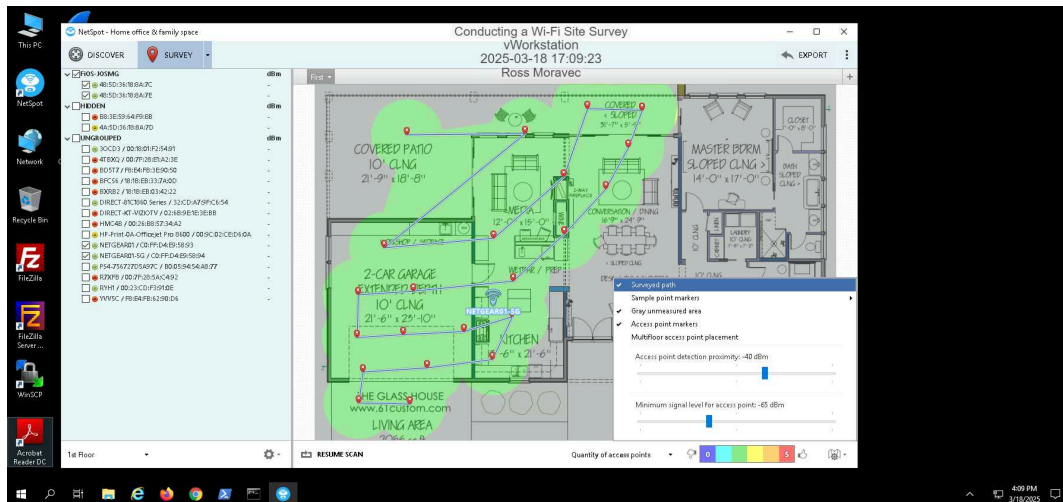
22. Make a screen capture showing the quantity of access points heatmap.



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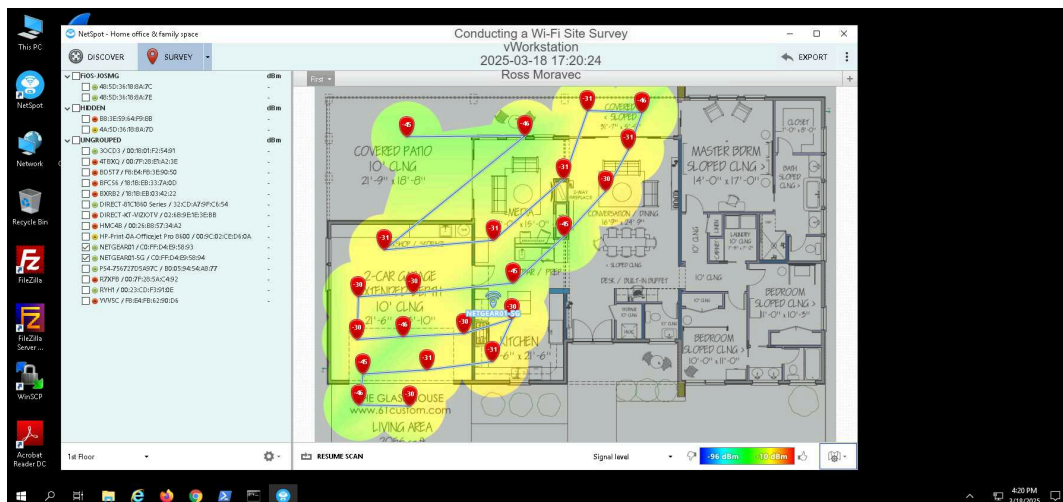
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25. Make a screen capture showing the updated quantity of access points heatmap.



Part 3: Identify Potential Sources of Interference

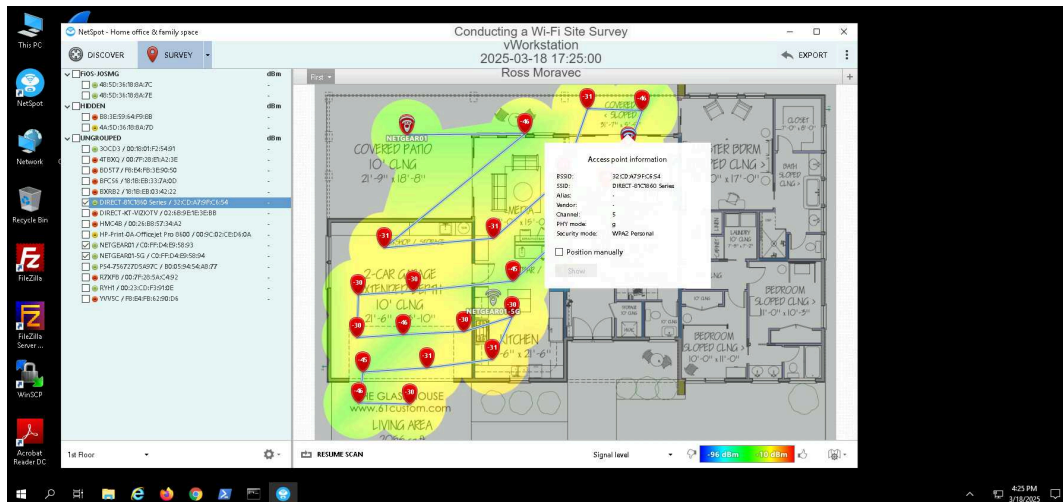
4. Make a screen capture showing the signal level heatmap with detailed sample points.



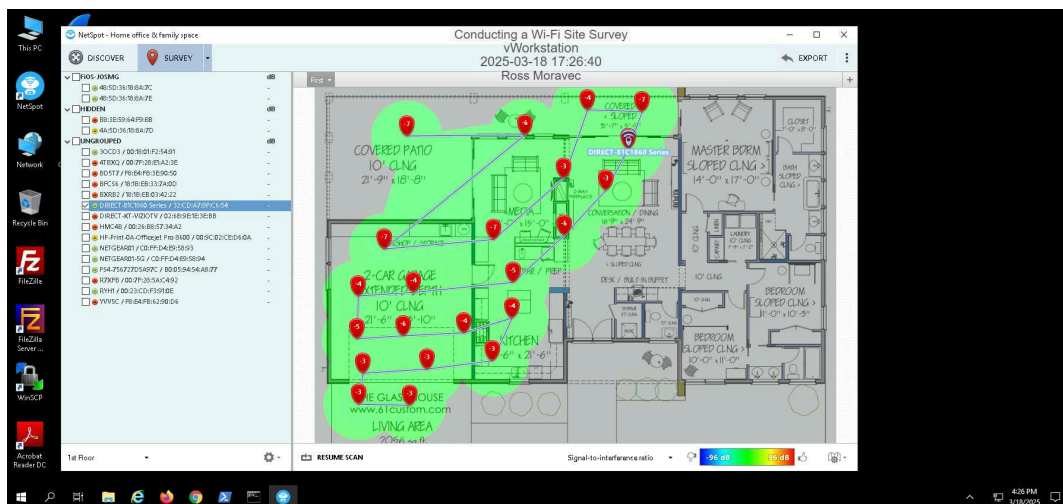
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9. Make a screen capture showing the **DIRECT-81C1860 Series** access point details.



12. Make a screen capture showing the **SIR values** for the **DIRECT-81C1860 Series** access point.



Challenge and Analysis

Part 1: Analyze Wireless Signal Strength and SIR

Document the BSSID of the AP you discovered nearest the low-signal sample point.

04:18:D6:B4:1C:70

Document the SIR value of the low-signal sample point you identified above.

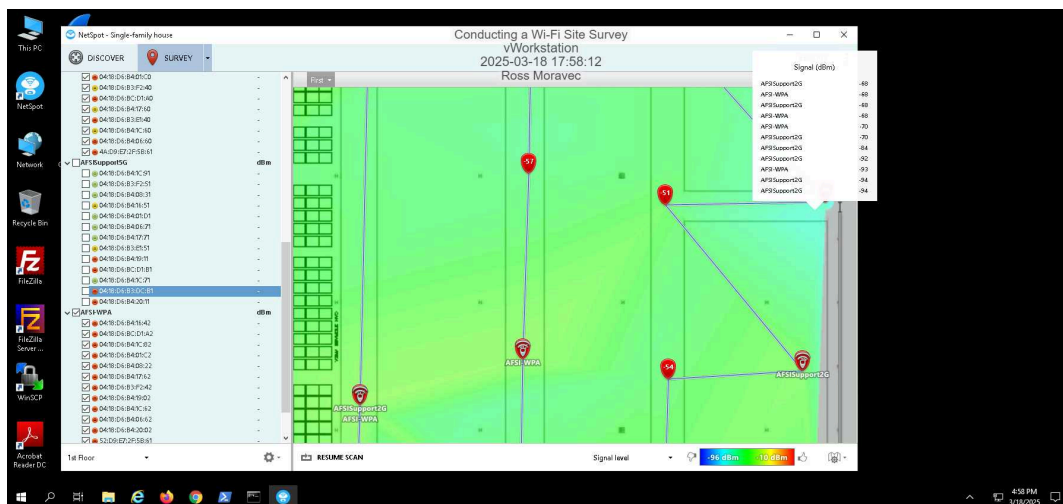
the sample point with about -70 dBm signal strength has an SIR of around -25 dB

Part 2: Analyze Wireless Network and Frequency Coverage

Document the network(s) with any dead zone(s).

BSSID SSID (if known) Signal (dBm) 84:78:AC:A1:B2:C3 AFSISupport2G -85 04:18:D6:B4:1C:70 AFSI-WPA -88 A0:21:B7:12:34:56 AFSISupport2G -90 2C:30:33:11:22:33 AFSISupport2G or AFSI-WPA -93

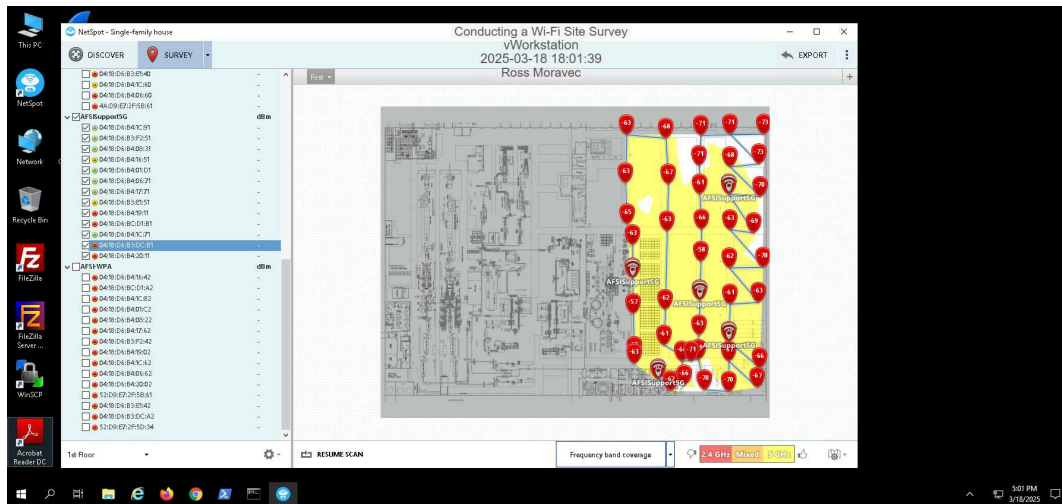
Make a screen capture showing the dead zone you identified above.



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Make a screen capture showing the 5GHz network coverage in the visualization.



Part 3: Find the Sweet Spot

Document which GM area has the better SIR on the guest network (using GMA1 or GMA2 to answer, per the screenshot below).

From the SIR values the right?hand General Manager's Area appears to have the higher peak SIR for the AFSI?GUEST network—specifically, one of the sample points reports an SIR as high as +13 dB. By contrast, the GM area on the left mostly hovers around +3 dB to +5 dB, with some negative entries as well. Because a higher positive SIR equates to better signal quality relative to interference, the GM area with sample point #10 (reaching +13 dB) is the better candidate for a new conference room.